

PASSWORD STRENGTH ANALYZER

JAVA MINI PROJECT

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INTRODUCTION

Password Strength Analyzer is a Java-based project used to check the strength of a password. It checks password length, uppercase letters, lowercase letters, numbers, and special characters. Based on these conditions, it calculates a score and displays whether the password is weak or strong. This project helps users create stronger passwords.

OBJECTIVES

The main objectives of the Password Strength Analyzer project are:

- 1. To check the strength of a password.**
- 2. To identify weak and strong passwords.**
- 3. To check password length.**
- 4. To check uppercase and lowercase letters.**
- 5. To check numbers and special characters.**
- 6. To calculate the password strength score.**
- 7. To help users create strong passwords.**

SOFTWARE REQUIREMENTS

Software Requirements:

- **Operating System: Windows**
- **Programming Language: Java**
- **IDE: Visual Studio Code**
- **JDK: Java Development Kit**
- **Input: User password**
- **Output: Password strength and score**

Hardware Requirements:

- **Computer or Laptop**
- **Minimum 4 GB RAM**
- **Keyboard**
- **Minimum 500 MB free storage**

PROJECT WORKING

The Password Strength Analyzer works in the following steps:

- 1. The program starts and asks the user to enter a password.**
- 2. The entered password is stored in a string variable.**
- 3. The program checks whether the password is common or not.**
- 4. It checks the password length.**
- 5. It checks uppercase letters, lowercase letters, numbers and special characters.**
- 6. The program calculates the password score based on the conditions.**
- 7. Finally, it displays the password strength as Weak or Strong.**

ALGORITHM

Step 1: Start the program.

Step 2: Import the Scanner class.

Step 3: Ask the user to enter a password.

Step 4: Read and store the password.

Step 5: Check whether the password is common.

Step 6: Check the password length.

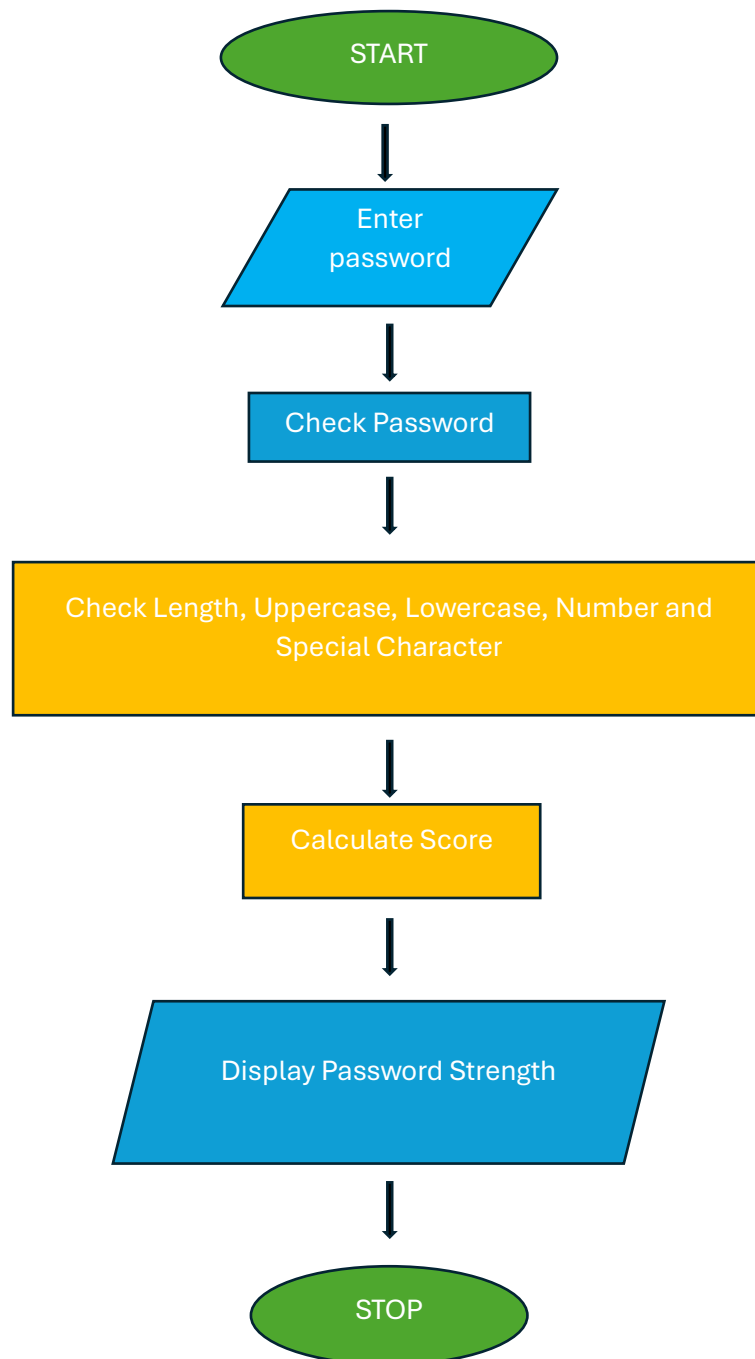
Step 7: Check uppercase letters, lowercase letters, numbers and special characters.

Step 8: Calculate the password strength score.

Step 9: Display the password strength.

Step 10: Stop the program.

FLOWCHART



SOURCE CODE

```
import java.util.Scanner;

public class PasswordStrengthAnalyzer {
    public static void main(String[] args) {
        System.out.println("=====");
        System.out.println("  PASSWORD STRENGTH ANALYZER");
        System.out.println("=====");

        Scanner sc = new Scanner(System.in);
        System.out.print("Enter your password: ");
        String password = sc.nextLine();
        String lowerPassword = password.toLowerCase();
        if (lowerPassword.equals("password") ||
            lowerPassword.equals("123456") ||
            lowerPassword.equals("qwerty") ||
            lowerPassword.equals("admin")) {
            System.out.println("Warning: This is a common password!");
        }

        System.out.println("Password entered: " + password);
        int score = 0;
        System.out.println();
        System.out.println("Password Analysis:");
        if (password.length() >= 8)
```

```
        System.out.println("✓ Length: Good");
    else
        System.out.println("X Length: Too Short");
    if (password.length() >= 8)
        score++;
    //Uppercase
    if (password.matches(".*[A-Z].*")) {
        score++;
        System.out.println("✓ Uppercase: Good");
    } else {
        System.out.println("X Uppercase: Missing");
    }
    //Lowercase
    if (password.matches(".*[a-z].*")) {
        score++;
        System.out.println("✓ Lowercase: Good");
    } else {
        System.out.println("X Lowercase: Missing");
    }
    //Number
    if (password.matches(".*[0-9].*")) {
        score++;
        System.out.println("✓ Number: Good");
    } else {
```

```
        System.out.println("X Number: Missing");
    }
    //Special character
    if (password.matches(".*[^a-zA-Z0-9].*")) {
        score++;
        System.out.println("✓ Special Character: Good");
    } else {
        System.out.println("X Special Character: Missing");
    }
    System.out.println("Score: " + score + "/5");
    if (score <= 2) {
        System.out.println("Password Strength: WEAK");
    } else if (score <= 4) {
        System.out.println("Password Strength: MEDIUM");
    } else {
        System.out.println("Password Strength: STRONG");
    }

    sc.close();

}

}
```

OUTPUT SCREENSHOTS

Output 1: Weak Password (Score: 1/5)

```
PS C:\Users\RAVI KIRAN\OneDrive\Desktop>Password Strength Analyser> c:: cd 'c:\Users\RAVI KIRAN\OneDrive\Desktop>Password Strength Analyser'; & 'C:\Program Files\Java\jdk-21.0.12.1\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:51840' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\RAVI KIRAN\AppData\Roaming\Code\User\workspaceStorage\6c9b275e4d1940abe5bd7fa49f0a6f04\redhat.java\jdt_ws>Password Strength Analyser_9c5d2a4a\bin' 'PasswordStrengthAnalyzer'

=====
          PASSWORD STRENGTH ANALYZER
=====

Enter your password: 123456
Warning: This is a common password!
Password entered: 123456

Password Analysis:
? Length: Too Short
? Uppercase: Missing
? Lowercase: Missing
? Number: Good
? Special Character: Missing
Score: 1/5
Password Strength: WEAK
PS C:\Users\RAVI KIRAN\OneDrive\Desktop>Password Strength Analyser> |
```

Output 2: Strong Password (Score: 5/5)

```
● PS C:\Users\RAVI KIRAN\OneDrive\Desktop>Password Strength Analyser> c:: cd 'c:\Users\RAVI KIRAN\OneDrive\Desktop>Password Strength Analyser'; & 'C:\Program Files\Java\jdk-21.0.12.1\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:57344' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\RAVI KIRAN\AppData\Roaming\Code\User\workspaceStorage\6c9b275e4d1940abe5bd7fa49f0a6f04\redhat.java\jdt_ws>Password Strength Analyser_9c5d2a4a\bin' 'PasswordStrengthAnalyzer'

=====
          PASSWORD STRENGTH ANALYZER
=====

Enter your password: Tiger@2026#Safe
Password entered: Tiger@2026#Safe

Password Analysis:
? Length: Good
? Uppercase: Good
? Lowercase: Good
? Number: Good
? Special Character: Good
Score: 5/5
Password Strength: STRONG
```

CONCLUSION

the strength of a password based on its length, The Password Strength Analyzer project was successfully developed using Java. It checks uppercase letters, lowercase letters, numbers, and special characters. The project calculates a score and displays whether the password is weak or strong. This project helps users understand the importance of creating strong passwords.

Thus, the project was completed successfully.